
UNIVERSITY OF THE PHILIPPINES DILIMAN
College of Science
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Physics 180 THV-1
Problem Set 1-2

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IV - BS Physics

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Problem 1.1

Go to **arXiv** and have a look at the relevant subjects for this course (nucl-ex, nucl-th, hep-ex, hep-ph, hep-th).

1. What do the first four digits of the **arXiv** number unique to a specific paper stand for?
 2. What are the **arXiv** numbers for the papers shown in Slides 22 and 23.
 3. Cite a single paper that fascinates you within the dates **August 11-20, 2026**, and briefly explain why you find this particular paper interesting.
-

Solution

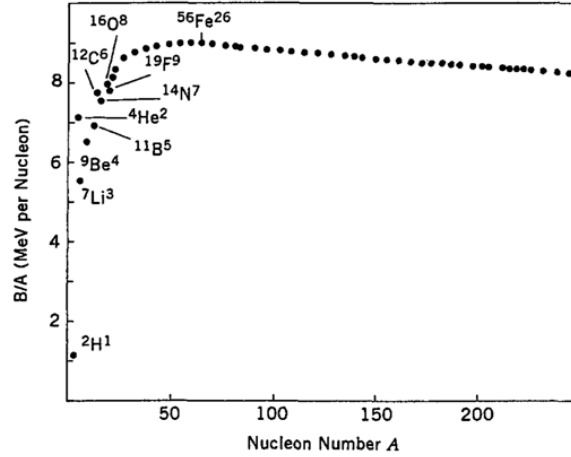
Problem 1.2

Go to InSpire and search for all papers from 1967. What is the most highly-cited paper from that year? Skim through its introduction. Why do you think is it highly cited?

Solution

Problem 2.1

The B/A plot looks like it peaks at a specific value of A . Using the curve approximated by SEMF, find the value of A that achieves this and find the peak value. What is the element that achieves this



Solution

From [1], we have the Bethe-Weizsäcker or SEMF formula.

$$B = a_1 A - a_2 A^{2/3} - a_3 \frac{Z^2}{A^{1/3}} - a_4 \frac{(Z - A/2)^2}{A} \pm a_5 A^{-1/2} \quad (3.1)$$

Dividing by A , we have

$$\frac{B}{A} = a_1 - a_2 A^{-1/3} - a_3 \frac{Z^2}{A^{4/3}} - a_4 \frac{(Z - A/2)^2}{A^2} \pm a_5 A^{-3/2} \quad (3.2)$$

We aim to maximize B/A with respect to A . Note, however that the function is both a function of A and Z , so the resulting equation will be dependent of Z . An important point here is that Z is dependent of A . Since we wish to know which isobar Z gives the most stable element, we must first maximize B/A with respect to Z . This is similar to the operation performed in class about finding N/Z , but we find the stability ratio for Z/A multiplied by A .

We find $Z(A)$ for which Z gives the most tightly bound isobar. Ignoring the pairing term for the smooth SEMF curve, we have

$$\frac{\partial B(Z, A)}{\partial Z} = 0 \quad (3.3)$$

Therefore, we have

$$\frac{\partial}{\partial Z} B = \frac{\partial}{\partial Z} \left(a_1 A - a_2 A^{2/3} - a_3 \frac{Z^2}{A^{1/3}} - a_4 \frac{(Z - A/2)^2}{A} \pm a_5 A^{-1/2} \right) \quad (3.4)$$

$$\frac{\partial}{\partial Z} B = -2a_3 \frac{Z}{A^{1/3}} - 2a_4 \frac{(Z - A/2)}{A} \quad (3.5)$$

$$0 = -2a_3 \frac{Z}{A^{1/3}} - 2a_4 \frac{(Z - A/2)}{A} \quad (3.6)$$

We multiply the expression with $-A/2$ to simplify

$$0 = a_3 Z A^{2/3} + a_4 \left(Z - \frac{A}{2} \right) \quad (3.7)$$

$$0 = a_3 Z A^{2/3} + a_4 Z - \frac{a_4 A}{2} \quad (3.8)$$

$$Z(a_3 A^{2/3} + a_4) = \frac{a_4 A}{2} \quad (3.9)$$

Thus, we have the most stable Z value for a given A

$$Z(A) = \frac{a_4 A}{2(a_3 A^{2/3} + a_4)} \quad (3.10)$$

We substitute this back to B/A .

$$\frac{B}{A} = a_1 - a_2 A^{-1/3} - a_3 \frac{Z(A)^2}{A^{4/3}} - a_4 \frac{(Z(A) - A/2)^2}{A^2} \pm a_5 A^{-3/2} \quad (3.11)$$

from this, we get

$$\frac{B}{A} = a_1 - a_2 A^{-1/3} - a_3 \frac{\left(\frac{a_4 A}{2(a_3 A^{2/3} + a_4)} \right)^2}{A^{4/3}} - a_4 \frac{\left(\frac{a_4 A}{2(a_3 A^{2/3} + a_4)} - A/2 \right)^2}{A^2} \pm a_5 A^{-3/2} \quad (3.12)$$

This simplifies to the following

$$\frac{B}{A} = a_1 - a_2 A^{-1/3} - a_3 \frac{a_4^2 A^{2/3}}{4(a_3 A^{2/3} + a_4)^2} - a_4 \frac{a_3^2 A^{4/3}}{4(a_3 A^{2/3} + a_4)^2} \pm a_5 A^{-3/2} \quad (3.13)$$

Combining the a_3 and a_4 terms, we have

$$-a_3 \frac{a_4^2 A^{2/3}}{4(a_3 A^{2/3} + a_4)^2} - a_4 \frac{a_3^2 A^{4/3}}{4(a_3 A^{2/3} + a_4)^2} = -\frac{a_3 a_4 A^{2/3} (a_4 + a_3 A^{2/3})}{4(a_3 A^{2/3} + a_4)^2} \quad (3.14)$$

$$= -\frac{a_3 a_4 A^{2/3}}{4(a_3 A^{2/3} + a_4)} \quad (3.15)$$

Hence,

$$\frac{B}{A} = a_1 - a_2 A^{-1/3} - \frac{a_3 a_4 A^{2/3}}{4(a_3 A^{2/3} + a_4)} \pm a_5 A^{-3/2} \quad (3.16)$$

Now, we maximize this in terms of A . We have¹

$$\frac{\partial}{\partial A} \left(\frac{B}{A} \right) = \frac{a_2}{3} A^{-4/3} - \frac{a_3 a_4}{4} \frac{\partial}{\partial A} \left[\frac{A^{2/3}}{a_3 A^{2/3} + a_4} \right] \mp \frac{3a_5}{2} A^{-5/2} \quad (3.17)$$

$$= \frac{a_2}{3} A^{-4/3} - \frac{a_3 a_4^2}{6} \frac{A^{-1/3}}{(a_3 A^{2/3} + a_4)^2} \mp \frac{3a_5}{2} A^{-5/2} \quad (3.18)$$

This maximum satisfies

$$\frac{a_2}{3} A^{-4/3} - \frac{a_3 a_4^2}{6} \frac{A^{-1/3}}{(a_3 A^{2/3} + a_4)^2} \mp \frac{3a_5}{2} A^{-5/2} = 0 \quad (3.19)$$

¹At this point I just used mathematica to solve for this shit

For a smooth SEMF curve, we usually drop the pairing term, giving us

$$\frac{a_2}{3}A^{-4/3} - \frac{a_3a_4^2}{6} \frac{A^{-1/3}}{(a_3A^{2/3} + a_4)^2} = 0 \quad (3.20)$$

This can then be further simplified by multiplying by $6A^{4/3}$

$$2a_2 - \frac{a_3a_4^2A}{(a_3A^{2/3} + a_4)^2} \quad (3.21)$$

Therefore

$$2a_2(a_3A^{2/3} + a_4)^2 = a_3a_4^2A \quad (3.22)$$

If we solve for A , we end up with this quartic equation if $A = x^3$

$$2a_2a_3^2A^{4/3} - a_3a_4^2A + 4a_2a_3a_4A^{2/3} + 2a_2a_4^2 = 0 \quad (3.23)$$

We solve for this numerically. I just used the Newton-Raphson zero finding algorithm. Code is provided on the appendices to show how that works.

Anyways, the resulting number for stable A , using the provided coefficients[1]

$$a_1 = 15.56 \text{ MeV}$$

$$a_2 = 17.23 \text{ MeV}$$

$$a_3 = 0.697 \text{ MeV}$$

$$a_4 = 93.14 \text{ MeV}$$

$$a_5 = 12 \text{ MeV}$$

We numerically solve for $A \approx 62$, see Appendices for the python code. Using the derived equation (3.10), we can calculate the most stable Z value for this A , and this results to $Z \approx 27$, or, **Nickel-62**.

$$N \approx 62 \quad (3.24)$$

$$Z \approx 27 \quad (3.25)$$

While I expected something akin to Iron-56, turns out, when you derive the actual energies, the actual answer is also Nickel-62, since it has the highest nuclear binding energy per nucleon. However, Iron-56 is more commonly called the most stable end-product of stellar nucleosynthesis because it has the lowest mass per nucleon and is produced in massive quantities by stars [2]

So hopefully this passes by the checker. Nickel and Iron are just beside each other on the periodic table.

References

- [1] M. Flores. Lecture Slides *Physics 180: Nuclear and Particle Physics*. Digital lecture slides, 2026. Available from course website.
- [2] Christopher S. Baird. What is the most stable nucleus? Science Questions with Surprising Answers, July 2024. Accessed: 16-08-2026.

Appendices

Code

```
1 import numpy as np
2
3 # SEMF coefficients from [1]
4 a1 = 15.56 # MeV
5 a2 = 17.23
6 a3 = 0.697
7 a4 = 93.14
8 a5 = 12.00
9
10 # Function definition f(A) (eq 3.23)
11 def f(A):
12     return 2*a2*a3**2 * A**(4/3) - a3*a4**2 * A + 4*a2*a3*a4 * A**(2/3)
13     + 2*a2*a4**2
14
15 # Function derivative f'(A)
16 def f_prime(A):
17     return (8/3)*a2*a3**2 * A**(1/3) - a3*a4**2 + (8/3)*a2*a3*a4 * A
18     **(-1/3)
19
20 # Newton-Raphson method
21 def newton_raphson(A0, tol=1e-12, max_iter=100):
22     A = A0
23     print(f"Iteration 0: A = {A:.10f}, f(A) = {f(A):.10e}")
24
25     for i in range(max_iter):
26         f_val = f(A)
27         f_prime_val = f_prime(A)
28
29         if abs(f_prime_val) < 1e-15:
30             print("Derivative too small, method failure.")
31             break
32
33         A_new = A - f_val / f_prime_val
34
35     print(f"Iteration {i+1}: A = {A_new:.10f}, f(A) = {f(A_new):.10e}")
36
37     if abs(A_new - A) < tol:
38         print(f"\nConverged after {i+1} iterations.")
39         print(f"Maximal value for      : A = {A_new:.10f}")
40         return A_new
```

```

39
40     A = A_new
41
42     print(f"\nMax iterations reached. A = {A:.10f}")
43     return A
44
45 # Solve for A
46 print("=" * 60)
47 print("Solving for A using Newton-Raphson method")
48 print("=" * 60)
49
50 A0 = 100 # Starting guess
51 print(f"\nStarting with A0 = {A0}")
52 print("-" * 60)
53 solution = newton_raphson(A0)
54
55 def Z(A):
56     num = a4 * A
57     den = 2 * (a3 * A**(2/3) + a4)
58     return num/den
59
60 print(f"\nMost stable element: Z = {Z(solution):.10f}")

```

Listing 1: Newton-Raphson solver script for A and Z

Output:

```

=====
Solving for A using Newton-Raphson method
=====

Starting with A0 = 100
-----
Iteration 0: A = 100.0000000000, f(A) = -2.0154515672e+05
Iteration 1: A = 61.9746717109, f(A) = -1.6138514552e+03
Iteration 2: A = 61.6645809668, f(A) = -1.7247545923e-01
Iteration 3: A = 61.6645478197, f(A) = -1.9790604711e-09
Iteration 4: A = 61.6645478197, f(A) = 0.0000000000e+00

Converged after 4 iterations.
Maximal value      : A = 61.6645478197
Most stable element: Z = 27.4076237435

```